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I pledge my honor that I have abided by the Stevens Honor System

Question 1:

```
int mystery1(int a, int b) {  
    int c = a - b,  
        d = (c >> 7) & 1,  
        mystery = a - c * d;  
    return mystery;  
}
```

Trace: mystery1(3, 7) returns _____

Trace: mystery1(8, 7) returns _____

Summary: _____

mystery1(3,7) returns 3

mystery(8,7) returns 8

Summary:

This function uses bitwise operations and returns the original value in all scenarios

```
int mystery2(int x) {  
    return (x && !(x & (x - 1)));  
}
```

Trace: mystery2(1) returns _____

Trace: mystery2(2) returns _____

Trace: mystery2(3) returns _____

Trace: mystery2(4) returns _____

Trace: mystery2(5) returns _____

Trace: mystery2(6) returns _____

Trace: mystery2(7) returns _____

Trace: mystery2(8) returns _____

Summary: _____

m2(1) returns 1

m2(2) returns 1

m2(3) returns 0

m2(4) returns 1

m2(5) returns 0

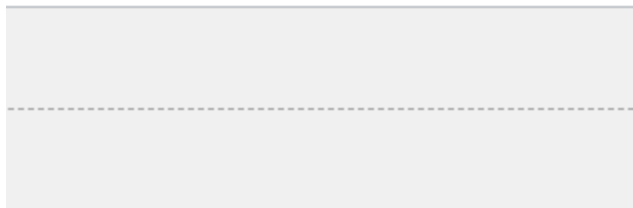
m2(6) returns 0

m2(7) returns 0

m2(8) returns 1

This algorithm checks if a number is a power of 2.

```
int mystery3(int x, int y) {  
    int s, c;  
    s = x ^ y;
```



```
    c = x & y;  
    while (c != 0) {  
        c = c << 1;  
        x = s;  
        y = c;  
        s = x ^ y;  
        c = x & y;  
    }  
    return s;  
}
```

Trace: mystery3(5, 7) returns _____

Trace: mystery3(2, 8) returns _____

Summary: _____

Initial Setup:

$x = 5$ (00000101)

$y = 7$ (00000111)

Iteration 1:

$s = x \oplus y = 00000101 \oplus 00000111 = 00000010$

$c = x \& y = 00000101 \& 00000111 = 00000101$

$c \neq 0$, enter loop:

$c = c \ll 1 = 00000101 \ll 1 = 00001010$

$x = s = 00000010$

$y = c = 00001010$

$s = x \oplus y = 00000010 \oplus 00001010 = 00001000$

$c = x \& y = 00000010 \& 00001010 = 00000010$

Iteration 2:

$c \neq 0$ (00000010), continue loop:

$c = c \ll 1 = 00000010 \ll 1 = 00000100$

$x = s = 00001000$

$y = c = 00000100$

$s = x \oplus y = 00001000 \oplus 00000100 = 00001100$

$c = x \& y = 00001000 \& 00000100 = 00000000$

Termination:

$c == 0$, loop terminates.

Returns $s = 00001100 = 12$

This one implements addition. Via leftwise bit operations and or operators.